Similar computations give

$$[\mathcal{L}_{\mathrm{A}}^{\dagger}, \mathcal{L}_{\mathrm{O}}^{\dagger}]\rho = \gamma p \rho_q + \sigma^2 \rho_{qp},$$
$$[\mathcal{L}_{\mathrm{B}}^{\dagger}, \mathcal{L}_{\mathrm{O}}^{\dagger}]\rho = \gamma U'(q) \rho_p.$$

To first order, the discretization map corresponding to scheme ⟦OBA⟧ alters the phase space density at each step as if it were solving the time dependent PDE

$$\rho_t = \hat{\mathcal{L}}^{\dagger} \rho,$$

where

$$\hat{\mathcal{L}}^{\dagger}\rho = \mathcal{L}_{\mathrm{LD}}^{\dagger}\rho + \frac{h}{2}\left[-pU''(q)\rho_p + U'(q)\rho_q + \gamma p \rho_q + \sigma^2 \rho_{qp} + \gamma U'(q)\rho_p\right] + \mathcal{O}(h^2).$$

The validity of this sort of expansion, even for small h, should be considered carefully. Defining

$$\mathcal{L}_{\mathrm{LD}}^{\dagger} = \mathcal{L}_{\mathrm{A}}^{\dagger} + \mathcal{L}_{\mathrm{B}}^{\dagger} + \mathcal{L}_{\mathrm{O}}^{\dagger}$$

as the Kolmogorov operator for Langevin dynamics, the assumption of validity becomes more credible as we find that the operator $\mathcal{L}_{\mathrm{LD}}^{\dagger}$ has a compact resolvent and therefore a discrete spectrum, i.e. simple eigenvalues, as discussed in the previous chapter. It will have one zero eigenvalue corresponding to the invariant density ρ_β and all the remaining eigenvalues lie in the negative half plane. Of course it is not just this operator whose properties are important, but also all the other operators arising in the expansion, and bounding these in relation to $\mathcal{L}_{\mathrm{LD}}^{\dagger}$ is not a simple task (particularly for higher orders).

7.7.3 Perturbation Technique

Suppose we are able to derive a series expansion for $\hat{\mathcal{L}}^{\dagger}$ of the form

$$\hat{\mathcal{L}}^{\dagger} = \mathcal{L}_{\mathrm{LD}}^{\dagger} + h\mathcal{L}_1^{\dagger} + h^2\mathcal{L}_2^{\dagger} + \mathcal{O}(h^3),$$

for some perturbation operators $\mathcal{L}_i^{\dagger}$. We assume that the invariant density has a similar expansion in powers of the stepsize, of the form given in (7.17). Inserting this ansatz into the stationary Fokker-Planck equation

$$\hat{\mathcal{L}}^{\dagger}\hat{\rho} = 0,$$